Project Design Phase-II

Data Flow Diagram & User Stories

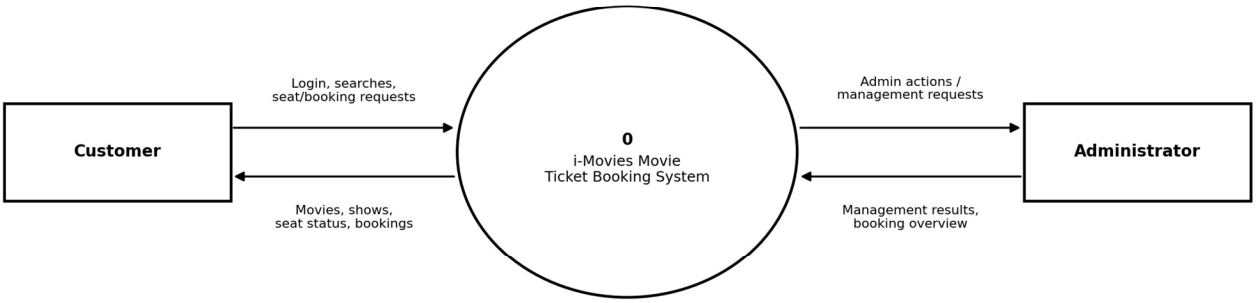
Date	22-10-2026
Team ID	
Project Name	i-Movies – Movie Ticket Booking System
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) illustrates the movement of information between external actors, application processes, and data stores. The diagrams below represent the documented i-Movies features and logical architecture.

DFD Level 0 — Context Diagram

i-Movies Movie Ticket Booking System — DFD Level 0 (Context Diagram)



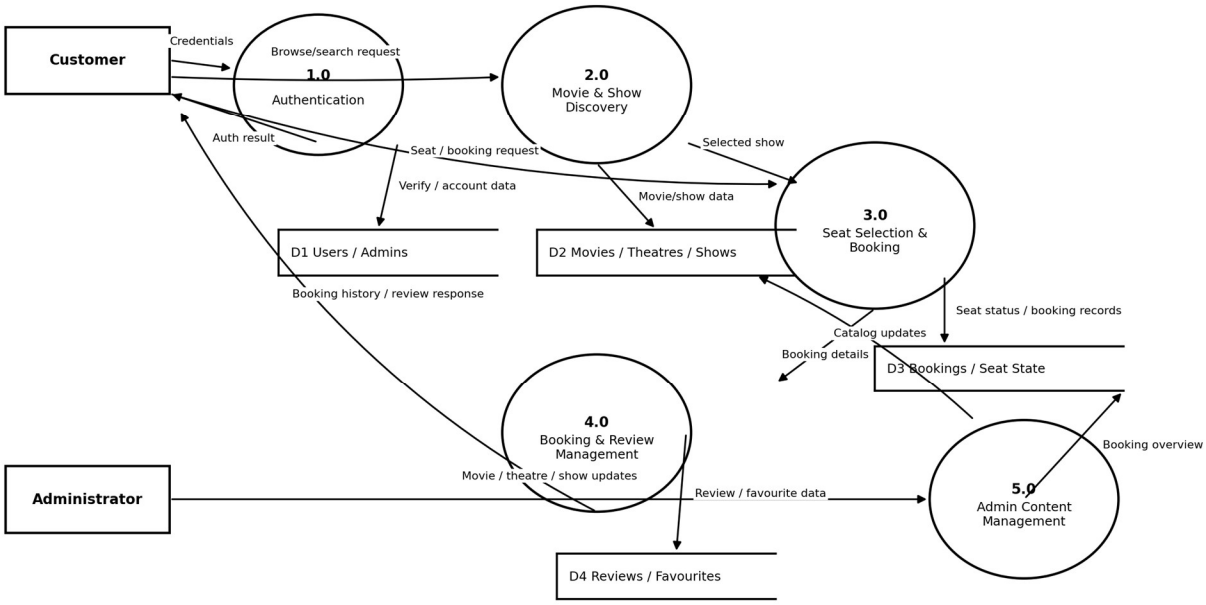
External actors exchange requests and responses with one system process.
Persistent data is managed internally by the application backend.

The customer sends authentication, discovery, seat-selection, and booking requests. The administrator sends content-management requests. The system returns relevant movie/show data, booking information, and administrative results.

DFD Level 1 — Main Processes and Data Stores

i-Movies Movie Ticket Booking System — DFD Level 1

Data stores represent MongoDB collections/models: User, Admin, Movie, Theatre, Show, Booking, Review, and Favorite. Arrows show the main logical data exchanges.



User Stories

User Story Table — i-Movies

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Authentication	As a customer, I can register and log in securely.	Valid credentials are accepted; invalid credentials are rejected.	High	Sprint 1
Customer	Profile Management	As a customer, I can view and update my profile.	Updated profile information is saved and shown.	Medium	Sprint 1
Customer	Movie Discovery	As a customer, I can browse, search, and filter movies.	Results reflect the search/filter criteria.	High	Sprint 1
Customer	Movie Details	As a customer, I can view movie details and available theatres.	Selected movie details and associated theatre/show information are displayed.	High	Sprint 1
Customer	Showtime & Seats	As a customer, I can view showtimes and seat availability.	The selected show displays current seat availability.	High	Sprint 2
Customer	Booking	As a customer, I can select available seats and create a booking.	Available seats are reserved; unavailable seats cause a conflict response.	High	Sprint 2
Customer	Booking Management	As a customer, I can view and cancel my bookings.	The customer can view their bookings; cancellation updates booking status and releases seats.	High	Sprint 2
Customer	Favourites	As a customer, I can save movies to my favourites.	A favourite can be added and is available in the customer's favourites list.	Medium	Sprint 3
Customer	Reviews & Ratings	As a customer, I can submit a movie review and rating.	A submitted review/rating is stored and associated with the movie/customer.	Medium	Sprint 3
Administrator	Content Management	As an administrator, I can create, update, and delete movies, theatres, and shows.	Authorized admin actions persist the changes.	High	Sprint 3
Administrator	Seat Pricing	As an administrator, I can configure seat tiers and prices.	Configured seat categories and prices are reflected in booking information.	High	Sprint 3
User	Preferences & Notifications	As a user, I can change the theme and receive in-app notifications.	Theme preference can be changed and notifications are displayed in the application.	Medium	Sprint 4